Functional Annotation of *moeA* (Q88L14 / PP_2123) in *Pseudomonas putida* KT2440

1. Summary — Answer to the Research Question

MoeA (Q88L14, locus PP_2123) is molybdopterin molybdenumtransferase (EC 2.10.1.1), the enzyme that catalyzes the final, metal-insertion step of molybdenum cofactor (Moco) biosynthesis. It ligates molybdenum into the dithiolene of molybdopterin (MPT) to produce the mature, active molybdenum cofactor. This reaction takes place in the **cytoplasm**, acting on an adenylylated molybdopterin (MPT-AMP) substrate provided by a MogA/MoaB-type adenylyltransferase partner, and delivering finished Moco for insertion into the cell's molybdoenzymes. In obligately aerobic KT2440 these clients are principally **xanthine dehydrogenase, aldehyde oxidoreductases, and formate dehydrogenase** (nitrate/DMSO/sulfite reductases are absent from this strain). MoeA is an ancient, deeply conserved enzyme, the bacterial ancestor of eukaryotic gephyrin/Cnx1.

The gene identity is unambiguous: the symbol *moeA*, the EC number, the MoeA protein family, and the domain architecture (MoaB/Mog-like domain IPR001453; MoeA-like IPR038987; MoeA_C_domain_IV IPR005111) all coincide with the canonical, well-characterized bacterial MoeA. Because MoeA is highly conserved and the *E. coli* ortholog is biochemically and structurally characterized, its function transfers to *P. putida* Q88L14 with high confidence.

2. Primary Function — the Catalyzed Reaction

Organism-specific annotation (UniProt Q88L14 / KEGG ppu:PP_2123): a 408-residue, single-chain enzyme; EC 2.10.1.1; KEGG ortholog **K03750** (molybdopterin molybdotransferase); orthology group **COG0303 (MoeA)**; FUNCTION "Catalyzes the insertion of molybdate into adenylated molybdopterin with the concomitant release of AMP"; **required cofactor Mg^{2+}**; pathway "Cofactor biosynthesis; molybdopterin biosynthesis." The recorded catalytic activity is:

> **adenylyl-molybdopterin (MPT-AMP) + molybdate (MoO_4^{2-}) → Mo-molybdopterin (Moco) + AMP + H^+** (Mg^{2+}-dependent)

MoeA performs this terminal metal-insertion step of Moco biosynthesis:

- In *E. coli*, **MoeA mediates the ligation of Mo to molybdopterin, while MogA enhances this process in an ATP-dependent manner** (Nichols & Rajagopalan 2007, P 17198377). MogA adenylylates MPT (forming MPT-AMP); MoeA then inserts molybdenum with concomitant release of AMP.
- This is the **final step of Moco biosynthesis: the incorporation of molybdenum into molybdopterin (MPT), the organic pyranopterin moiety of Moco** (Xiang et al. 2001, P 11525167).
- The same reaction — **transfer and insertion of Mo into MPT** — is carried out in eukaryotes by the two-domain proteins Cnx1 (plants) and gephyrin (mammals), whose E-domain is the MoeA homolog (Schwarz et al. 2001, P 11554796).

Substrate specificity: the organic substrate is molybdopterin / adenylated molybdopterin (MPT-AMP); the metal substrate is molybdate; Mg^{2+} is required as a catalytic cofactor. Notably, MoeA-family proteins can also handle tungstate: in tungsten-utilizing organisms, specialized MoeA paralogs dedicate biosynthesis toward tungsten cofactors (Debnar-Daumler et al. 2014, P 24214948), underscoring that the enzyme's role is metal insertion into the pterin dithiolene.

3. Structure–Function Basis

- The *E. coli* MoeA monomer is **highly elongated with four clearly separated domains, one of which is structurally related to MogA** (Xiang et al. 2001, P 11525167; Schrag et al. 2001, P 11428898).

Q88L14's InterPro domains (MoaB/Mog domain; MoeA-like; MoeA C-domain IV) match this architecture.

- **The active form of MoeA is a dimer**, with the putative active site in **a cleft formed between domain II of one monomer and domains III and IV of the second monomer** (Xiang et al. 2001, P 11525167) — i.e., a composite, inter-subunit active site requiring homodimerization.
- Mutational analysis maps two functional activities to the active-site cleft, and a variant's Moco-binding ability correlates with activity (Nichols & Rajagopalan 2007, P 17198377). Conserved acidic residues are essential: **replacements such as Asp259Ala abolished MoeA function in Mo-molybdopterin formation and stabilization, and the inability to restore nitrate reductase activity** (Sandu & Brandsch 2002, P 11934270).

4. Localization

Moco biosynthesis is a **cytosolic** process; MoeA is a soluble cytoplasmic enzyme (no signal peptide or transmembrane segments; consistent with UniProt annotation). It carries out metal insertion in the cytoplasm and the resulting Moco is subsequently inserted into apo-molybdoenzymes, some of which are then exported (e.g., periplasmic nitrate reductase NapA) or remain cytoplasmic.

5. Pathway Context and Physiological Role

MoeA operates at the **convergence of two branches** of Moco biosynthesis: the pterin branch (MoaA/MoaC → cyclic pyranopterin monophosphate; MoaD/MoaE + MoeB → MPT) and the metal-incorporation branch (MogA/MoaB adenylylation of MPT + MoeA Mo insertion; downstream MobA/MobB further modify Mo-MPT to the molybdopterin-guanine dinucleotide

(MGD/bis-MGD) form). Its product Moco is **required for all molybdoenzymes except nitrogenase** (Schwarz et al. 2001, P 11554796).

KT2440 encodes a complete Moco pathway (KEGG mapping): MoaA (PP_1969, PP_2482, PP_4597), MoaC (PP_1292), MoaB/MogA-like scaffolds (PP_1293; and MoaB paralogs PP_2122, PP_4600), MoeB sulfurtransferase (PP_1294), **MoeA (PP_2123, the target, K03750)**, and downstream MobA (PP_4003) and MobB (PP_3309) for MGD synthesis. Notably a dedicated *mogA* gene (K03831) is absent — the adenylyltransferase partner activity is supplied by MoaB/MogA paralogs, one of which (PP_2122) is operon-coupled to *moeA*. The presence of MobA/MobB confirms that MoeA-derived Mo-MPT is matured to the MGD form used by the Gram-negative DMSO-reductase-family molybdoenzymes.

Physiological evidence from bacteria (illustrating MoeA's essentiality for molybdoenzyme function; note these particular clients differ by organism — see the KT2440-specific survey below): - In *Burkholderia thailandensis*, a **moeA transposon mutant could not grow anaerobically with nitrate as sole terminal electron acceptor**, with defects in nitrate reductase activity, biofilm formation, and motility, all restored (except motility) by complementation (Andreae et al. 2014, P 24239959).
- The rice MoeA homolog OsCNX1, when restored, recovered **MoCo-dependent enzyme activities such as xanthine dehydrogenase, aldehyde oxidase, nitrate reductase and sulfite oxidase** (Liu et al. 2019, P 30471121), illustrating the breadth of downstream molybdoenzymes.

Genomic context in KT2440: *moeA* (PP_2123, 2,421,713–2,422,939, forward strand) lies immediately downstream of and **overlaps by ~16 bp** with PP_2122 ("molybdenum cofactor biosynthesis protein B," a MogA/MoaB-family gene), indicating translational coupling / co-transcription. This physical pairing mirrors the functional partnership between MoeA and its MogA/MoaB-type adenylyltransferase partner (which generates the MPT-AMP substrate), supporting coordinated expression of the final metal-insertion module.

Molybdoenzyme clients specific to KT2440 (KEGG survey): Importantly, obligately aerobic *P. putida* KT2440 **lacks** the nitrate reductases (respiratory NarG, assimilatory, periplasmic NapA), DMSO/TMAO reductase, biotin sulfoxide reductase, and sulfite oxidase that are the prominent Moco clients in facultative anaerobes such as *E. coli* and *Burkholderia*. Instead, its Moco-dependent enzymes are: **xanthine dehydrogenase** (XdhA, PP_3308, with small subunit PP_3310 — flanking MobB PP_3309, i.e., a purine-catabolism/molybdo gene cluster), **aldehyde oxidoreductases** (PaoC-type, PP_2477, PP_3621), and **formate dehydrogenase** (PP_0489, PP_2185). Thus in KT2440, MoeA-derived Moco gates **aerobic purine catabolism, aldehyde oxidation, and formate oxidation** rather than anaerobic nitrate/S-oxide respiration. Consistent with the research brief's emphasis on the precise role rather than pleiotropy, MoeA's effects are pleiotropic only because it is the single gate for cofactor maturation; its own biochemical act is the specific, defined metal-insertion reaction.

6. Evolutionary Evidence

MoeA **exists in all domains of life** and is the ancestor of eukaryotic gephyrin (mammals) and Cnx1 (plants), where MogA and MoeA are fused (Xiang et al. 2001, P 11525167). Phylogenetic and structural analysis confirms MoeA is an **essential enzyme for Moco biosynthesis**; early eukaryotes acquired MoeA from Bacteria, MogA–MoeA fusion arose in the opisthokont ancestor, and gephyrin later gained receptor-clustering roles at inhibitory synapses (Megrian et al. 2025, P 39809875). The *P. putida* protein represents the ancestral, catalytically dedicated bacterial form without the eukaryotic scaffolding functions.

7. Supported and Refuted Hypotheses

Supported - H1: Q88L14/MoeA catalyzes molybdenum insertion into molybdopterin (final Moco step). — Strongly supported by structural, biochemical, and mutational data on orthologs. - H2: MoeA functions as a homodimer with an inter-subunit active-site cleft and conserved catalytic Asp residues. — Supported by crystallography and mutagenesis. - H3: MoeA acts cytoplasmically and its product Moco is essential for the strain's molybdoenzymes. — Supported by mutant phenotypes across taxa; in KT2440 specifically the clients are xanthine dehydrogenase, aldehyde oxidoreductases, and formate dehydrogenase (KEGG survey), not nitrate/DMSO reductases. - H4 (organism-specific): KT2440's Moco demand reflects an aerobic lifestyle. — Supported: the strain lacks anaerobic-respiration molybdoenzymes, so MoeA gates aerobic purine/aldehyde/formate oxidation.

Not applicable / Refuted - The *moeA* symbol is NOT ambiguous here; it is not a transporter, structural protein, or signaling molecule. It is a biosynthetic transferase enzyme. - MoeA is not itself a molybdoenzyme (it does not use Moco); it makes Moco.

8. Limitations and Future Directions

- No *P. putida* KT2440-specific biochemical or structural study of PP_2123 was found; functional assignment relies on high-confidence orthology to *E. coli* MoeA and conserved domain architecture. Direct enzymatic characterization or a KT2440 knockout phenotype would confirm substrate specificity (Mo vs W) and downstream molybdoenzyme dependencies in this organism.
- The exact kinetic mechanism (order of Mo insertion vs AMP release; Mg^{2+} requirement) is inferred from ortholog studies.
- Whether *P. putida* MoeA shows any tungstate handling or paralog specialization is untested.
- The molybdoenzyme inventory was derived from KEGG KO markers, which could miss divergent or unannotated enzymes; nonetheless the clear absence of nitrate/DMSO/sulfite-reductase KOs is consistent with KT2440's known obligately aerobic, non-denitrifying physiology.

Citations: PMIDs 17198377, 11525167, 11428898, 11934270, 11554796, 39809875, 24239959, 30471121, 24214948.
